

EVISTREAMS: Human-in-the-Loop AI Data Extraction for Systematic Reviews in Medicine*

Sai Karthik Kosuri¹ Ankita Shashikant Bhosale¹ Michael Glick¹
Alonso Carrasco-Labra¹ Chris Callison-Burch²

¹Center for Integrative Global Oral Health (CIGOH), Penn Dental Medicine, University of Pennsylvania

²Department of Computer and Information Science, University of Pennsylvania

Corresponding author: karthik9@upenn.edu

Abstract

Systematic reviews underpin clinical guidelines, yet their data-extraction step is a major expert-labor bottleneck bound by a protocolized workflow: two reviewers extract each study independently, an adjudicator resolves disagreements, and the team keeps an auditable record of how every value was produced. Large language models can assist with extraction, but that assistance must fit established review protocols and preserve reproducibility. We present EVISTREAMS, a live, open-source, no-code web platform that puts review teams in control of AI-assisted extraction at three key stages: *program design* (a structured decomposition approved before any code runs), *field specification* (typed field definitions calibrated from a pilot), and *extracted predictions* (reviewer-blinded dual review with adjudication). Working through a form builder, a domain expert defines typed fields rather than prompts, runs extraction over uploaded PDFs, inspects every value alongside the supporting passage it came from, and resolves a reviewer-blinded dual review into an auditable consensus export. An evaluation across four clinical corpora and three frontier model families, released with the system, shows that extraction quality is shaped far more by the field specification than by the choice of model. EVISTREAMS is live at <https://evistreams.com/demo> and released under Apache-2.0.

1 Introduction

Systematic reviews inform clinical and public health practice guidelines, health-technology assessments, and payer and regulatory decisions. Their authority rests on a demanding protocol: two reviewers extract each study independently, an adjudicator resolves disagreements, blinding guards against anchoring, and the team keeps an auditable record of how every value was produced. Data extraction (the transfer of numerical and categorical

findings into structured tables) is among the most labor-intensive steps, consuming hundreds of expert hours per review (Higgins et al., 2023). Large language models can populate an extraction form accurately (Simmons et al., 2025), but an extraction error can be catastrophic: a single mis-extracted sample size or effect estimate can propagate into a pooled estimate and a clinical recommendation. Therefore we are motivated to design a human-in-the-loop system for systematic reviews that lets humans oversee the review, trace each value, and verify the outputs. We investigate three key stages that put humans in control of AI-generated extraction while meeting the evidentiary standards of the discipline.

Our main finding is that extraction quality is governed far less by *which* model reads a paper than by *how* the task is specified. This echoes the growing observation in clinical information extraction that “success may increasingly hinge on the clear articulation of objectives, rather than on singular model performance” (Hein et al., 2025), and that customized, domain-tuned prompts dominate generic ones (Li et al., 2025), which we elevate into an architectural commitment. We study extraction across four clinical corpora. Choosing the model or changing the extraction pipeline changes performance by only a few F1 points, with mixed direction across corpora. The field specification has the largest effect: a richer, more detailed specification produces better results.

We present EVISTREAMS (short for evidence streams), an open-source, live platform running at <https://evistreams.com/demo> that puts domain experts in control of AI-assisted extraction through a no-code interface. Human review enters at three key stages: (1) *program*, where the system proposes a plan for extracting each field from the PDF and a reviewer approves, edits, or rejects it before any code runs; (2) *specification*, where each field’s definition is tested on a handful of papers

*Live demo: <https://evistreams.com/demo>

FIELD NAME
surgery_type

TYPE
text

DESCRIPTION

Type of surgery the children underwent, verbatim from the paper (e.g. 'tonsillectomy with/without adenoidectomy', 'dental extraction', 'orthodontic separator placement', 'lingual herniorrhaphy'). Use 'NR' if not stated.

EXTRACTION HINTS

Where or how to find this value — e.g. "look in the Methods section". Soft guidance for the AI.

→ Look in the methods, participants, or procedure section describing the surgical intervention

+ Add hint

RULES

Hard constraints the AI must follow — e.g. "return NR if not found", "never infer". Rules override hints.

• Extract the surgery/procedure name verbatim as written in the document

• Preserve qualifiers such as 'with or without adenoidectomy' if present

+ Add rule

EXAMPLES

VALUE
tonsillectomy with adenoidectomy

+ Add example

Figure 1: A structured field in the form builder (surgery_type): description, extraction hints, rules, and examples, edited directly by the reviewer.

and the reviewer refines it based on what comes back; and (3) *predictions*, where every extracted value is checked by two independent reviewers, and a third resolves any disagreement. Figure 1 shows how the domain expert controls extraction by adding hints, rules, and examples.

Every AI-extracted value is paired with the verbatim quote it was extracted from, which reviewers can inspect. Unlike prior work, which lets users correct only the final output, EVISTREAMS gives many ways to control how the system behaves: from the extraction plan to the field specification to the final adjudication step.

2 The EVISTREAMS System

We designed EVISTREAMS for the ADA Living Guidelines Program, a collaboration between the American Dental Association and our lab, the Center for Integrative Global Oral Health (CIGOH), that keeps oral health guidelines updated as new evidence comes in (Carrasco-Labra et al., 2026). Our team uses EVISTREAMS end to end: a methodologist designs a structured extraction form and approves the pipeline the system proposes for it, then reviewers run extraction over the uploaded PDFs and adjudicate the results, with human review entering at three key stages (Figure 5).

2.1 Extraction pipeline

A methodologist designs a *form*, giving only a name, a description, and optionally an example for each field. The system groups related fields into stages and proposes this grouping as a plan (Fig-

ure 2) for the reviewer to approve: never executable code. Once approved, EVISTREAMS¹ adds hints and rules to each field and turns the plan into extraction code automatically: a fixed compiler maps the approved plan to predefined DSPy components; no LLM writes this code. Separately, uploaded PDFs are stored in S3 and converted to Markdown using Datalab (Datalab, 2024).

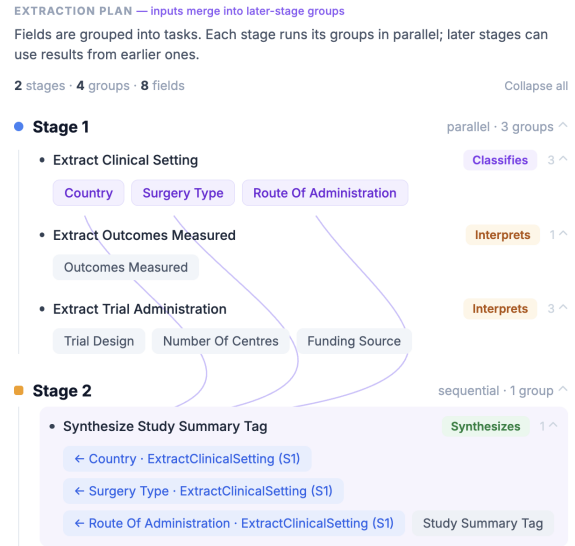


Figure 2: An extraction plan: fields are grouped into tasks, and each stage runs its groups in parallel; later stages can use results from earlier ones. Here Stage 1 extracts clinical setting, outcomes, and trial administration fields in parallel; Stage 2 synthesizes a study summary tag from three of Stage 1’s outputs.

Once fields are enriched, each group from the plan becomes a DSPy signature (Khattab et al., 2024): fields in the same group share one Chain-of-Thought signature (Wei et al., 2022), and independent groups run at the same time. A DSPy module then runs each signature to extract the data. We skip DSPy’s automatic prompt optimizers (Opsahl-Ong et al., 2024; Agrawal et al., 2025): at our scale (5–20 example papers), we prioritize direct expert editing over automatic prompt optimization (Li et al., 2025). Every value comes back with its source text, in the same format across every model (Figure 3).

2.2 Structured field specification

Each field’s specification has four elements (Figure 1, from the introduction): a **description** of

¹Backend: https://github.com/karthik-strikes/evistream_backend. Frontend: https://github.com/karthik-strikes/evistream_frontend. Released under Apache-2.0.

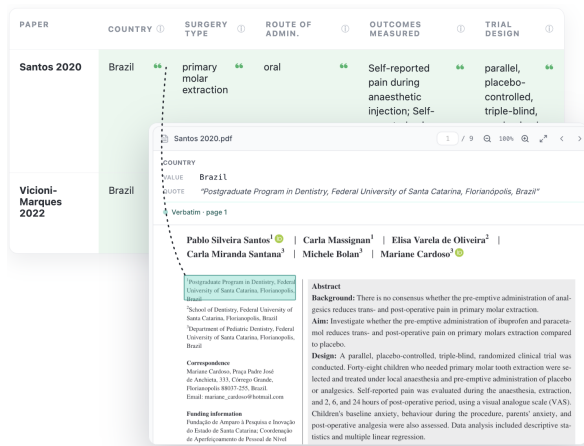


Figure 3: Run: every extracted value can be expanded to its verbatim source quote and page in the original PDF, so a reviewer can verify a value (here country: “Brazil”) against exactly the sentence it came from.

the concept (always present, user-owned and never rewritten); **hints** locating the evidence within a paper; **rules** imposing hard output constraints (e.g. “return the analyzed N , not the enrolled N ”); and **examples** anchoring the expected value shape. Hints and rules are filled in automatically: a LangGraph (LangChain, 2024) chain runs a prompt over the field’s description and any user-given examples to generate them, consistent with what the user wrote.

Splitting *what* a field means from *how* to find it lets a methodologist localize an error to just one element. These elements are also what pilot calibration edits later (§2.4); the reviewer refines them after seeing pilot results.

2.3 Table extraction

Medical-study tables record one row per **arm** (the different drugs or interventions a study tests for a condition, e.g., ibuprofen versus an alternative drug), or one row per (arm, follow-up, outcome) triplet. Naive LLM extraction on these tables often truncates the table, drifts in meaning, or hallucinates columns that were never asked for.

We avoid this with content anchoring (Khot et al., 2023): instead of reading a table positionally, we first extract only a few *anchor columns* that name each row (e.g., drug name and dose), then fan out one call per row to fill in the rest. For example, a 10-column table repeated across 400 papers easily loses columns under naive extraction; anchoring each row to its drug and dose first, then filling in the remaining cells per row, is designed to reduce row and column drift, since each call already knows

which row it belongs to.

2.4 Human review at three key stages

Program review (§2.1, Figure 2) is implemented as a LangGraph (LangChain, 2024) workflow: the proposed plan is a directed acyclic graph (DAG) of extraction sub-tasks with field-level dependencies, validated mechanically for cycles and for missing or duplicated field assignments (regenerating with structured feedback on failure), and paused at an `interrupt_before` checkpoint until the reviewer approves, edits, or rejects it.

Specification calibration occurs after a form is active: the reviewer tests the form on a handful of papers (typically 3–5), sees the results, and re-edits the fields (Figure 1) based on what comes back. These corrections update the corresponding signatures at runtime: any change to a field’s hints, rules, or description is reflected directly in its prompt.

Figure 4: Adjudication: fields where the AI and both reviewers already agree are marked *Agreed*; fields with a disagreement (here `trial_design` and `number_of_centres`) show the AI’s and both reviewers’ entries side by side for the adjudicator to pick or correct.

Adjudication² occurs after extraction: two reviewers independently complete the same form on the same paper under reviewer blinding (each sees the AI pre-fill but not the other’s entries). Field-level disagreements are computed on demand, and an adjudicator resolves each conflict (Figure 4). For every field, the system links the AI’s pre-fill,

²Screencast (≤ 2.5 min): <https://www.youtube.com/watch?v=j1Gxyu0KitU>.

both reviewers’ entries, and the adjudicator’s decision together, so the full history of how a value moved from AI guess to final answer can be looked up later. Reviewer blinding keeps one reviewer’s entry from anchoring the other, preserving the independent review that systematic-review protocols require; it does not blind either reviewer to the AI pre-fill itself.

Implementation. EVISTREAMS is a Next.js/FastAPI stack with Celery workers, Redis-backed live WebSocket progress, Supabase (Postgres) storage, Markdown ingestion (Datalab), and a multi-model fallback chain over the three frontier families.

3 Evaluation

We evaluate EVISTREAMS on four clinical datasets, measure the differences among three frontier models, and analyze the errors that occur. The choice of frontier model does not dramatically affect quality: all three perform similarly. The field specification drives the largest variance in quality.

Setup. We score results with a field-typed harness, released publicly alongside EVISTREAMS and implemented independently from the extraction pipeline to reduce evaluation coupling. We use four clinical corpora, each from a completed systematic review with expert-prepared ground truth: **oral cancer** (internal dataset; 43 diagnostic-accuracy studies, four forms, 47 fields; primary), **antibiotic prophylaxis** (CD010266 (Brignardello-Petersen et al., 2015); 10 trials), **periodontitis** (CD004714 (Simpson et al., 2022); 29 trials, outcomes per arm $\times$ time point), and **ibuprofen** (CD015432 (Pessano et al., 2024); 43 trials, a fourth-domain check); all three model families were run on every corpus (Appendix A)³.

Each field is scored by a type-matched strategy, and record-level forms are aligned by a globally optimal Hungarian assignment over row-identity keys before scoring, so a missed arm counts against recall and a spurious one is an over-extraction. We report micro- and macro-F1, which agree within ~ 0.02 on most forms (up to ~ 0.06 on the hardest oral-cancer forms).

³Extraction models: Claude Sonnet 4.6 (Anthropic, 2026), GPT-5.5 (OpenAI, 2026), and Gemini 3.1 Pro (Google DeepMind, 2026) (preview); the free-text judge is Claude Sonnet 4.6. We score every study whose full-text PDF was available: 10 of 11 antibiotic-prophylaxis trials and 29 of 35 periodontitis studies (the rest were excluded for PDF unavailability); ibuprofen is complete at 43 of 43.

Corpus	Studies	Gemini	GPT	Claude
Ibuprofen	43	0.842	0.845	0.851
Oral cancer	43	0.741	0.758	0.763
Antibiotic	10	0.896	0.851	0.898
Periodontitis	29	0.827	0.799	0.846

Table 1: Overall micro-F1 per corpus, three frontier model families through the same pipeline with identical form specifications (best per row in bold). Macro- and micro-F1 differ by at most 0.03 at the corpus level; per-form macro-F1 detail in Appendix B.

We test the model, the pipeline decomposition, and the field specification one at a time, holding the other two fixed. We score against the expert-prepared reference data, though this ground truth can itself contain reviewer conventions or occasional errors. The free-text judge (Claude Sonnet 4.6) is itself one of the three extraction models, a potential circularity readers should weigh when interpreting free-text scores.

3.1 Production quality and robustness

Table 1 reports overall micro-F1 per corpus for all three frontier families through the identical pipeline and specifications (per-form detail in Appendix B). Quality varies little across the families tested (Claude and Gemini within 0.03 micro-F1 on every corpus, GPT trailing by up to ~ 0.05 on antibiotic and periodontitis), and is bounded far more by the form and its source text than by the model (the between-corpus range, 0.741–0.898, dwarfs the between-model one). Oral cancer, our most complex corpus (four forms, 47 fields), is consistently hardest across all three models; antibiotic prophylaxis is consistently easiest, tracking form complexity rather than any one model’s weakness. With specification and pipeline fixed, the provider can thus be chosen on cost, latency, or governance rather than accuracy. Collapsing the multi-signature pipeline into a single hand-tuned prompt changes overall F1 by at most 0.036 with mixed direction (the one sizable swing, antibiotic tables, is GPT-specific: $\Delta = -0.096$ vs. -0.005 for Claude and Gemini), so decomposition provides governable structure, not accuracy gains, a robustness pattern independently corroborated by concurrent work (Aal Abdulsalam et al., 2026; Hart and Bergamaschi, 2026) (Appendix D).

Discrepancy type	Share
Methodological (ground-truth/protocol side)	68%
Genuine model errors (30%)	
Missing record or outcome row	7%
Stated-value omission (false NR)	6%
Wrong arm, table, or time point	6%
Cohort or adjacent-column confusion	5%
Denominator or document-context error	4%
Arm or regimen conflation	1%
Uncertain/unadjudicable	2%

Table 2: Taxonomy of 1,003 reviewed discrepancies from an error-enriched sample of 52 studies. Percentages are shares of reviewed discrepancies (rounded; subtypes need not sum to the group total), not extraction error rates. Detailed definitions and causes appear in Appendix E (Table 6).

3.2 Field specification is the largest observed lever

Across the four calibrated corpora and three tested model families, richer field specifications improved macro-F1 in 40 of 51 model-form comparisons (78%), with a mean ΔF_1 of 0.049 and a median of 0.014 (full per-form ablation in Appendix C). The mean is strongly influenced by the oral-cancer index_test form: removing hints, rules, and examples reduced its macro-F1 from 0.724 to 0.247 for Claude, with similarly large declines for Gemini and GPT. Excluding this form, the mean improvement was 0.023, and the full specification remained better in 37 of 48 comparisons (77%). Thus, richer specifications usually provide modest gains but can become decisive for fields requiring disambiguation, source localization, or non-obvious extraction conventions. Model and decomposition effects, reported above, were comparatively small and mixed in direction.

3.3 Error profile

We reviewed and classified every scored discrepancy in 52 studies (all 10 antibiotic trials; 15/43 oral-cancer, 15/43 ibuprofen, 12/29 periodontitis, each of the latter three by set-cover over the lowest-scoring fields): an LLM-assisted first-pass classification, followed by manual verification of every label by an expert systematic-review methodologist. Table 2 summarizes the discrepancy taxonomy and prevalence: of 1,003 discrepancies, two-thirds are **methodological** (ground-truth, convention, ambiguity, or scoring-side), not model failures; under a third are **genuine model errors**, concentrated in

dense multi-arm, multi-table reporting; the rest are unadjudicable. Among genuine model errors, omissions and context-assignment mistakes were both common. Missing rows and false-NR omissions accounted for 138 discrepancies, while 160 involved incorrect arms, time points, cohorts, denominators, or regimens. Pilot calibration is intended to reduce specification-driven misses, while dual review and adjudication provide a final check on residual errors. Detailed definitions and causes for each subtype are provided in Appendix E.

4 Demonstration

Demo walkthrough. In the reviewer demo, a methodologist first uploads a batch of PDFs, which EVISTREAMS converts to Markdown via Datalab (Datalab, 2024) in the background while the methodologist keeps working. The methodologist then creates a form, naming each field with a description and an optional example; EVISTREAMS proposes a decomposition that groups related fields into parallel extraction stages (Figure 2), and the methodologist reviews and approves the plan before any code runs.

EVISTREAMS then runs a pilot study, extracting the form over 3–5 papers so the methodologist can check each AI value against the expected value and refine hints, rules, or examples on any field that came back wrong; edits save live and are reflected in the next pilot run.

Once the pilot output matches expectations, the methodologist runs the approved form over the remaining documents, and every extracted value is shown alongside the supporting passage it was drawn from (Figure 3). Two reviewers then carry out manual extraction: each independently completes the same form on the same paper under reviewer blinding, accepting or correcting the AI pre-fill. EVISTREAMS surfaces their field-level disagreements, and an adjudicator resolves each conflict into the consensus export (Figure 4), which records the AI suggestion, both reviewer entries, the adjudicated value, and inter-rater agreement.

Target audience. Systematic-review teams: clinical experts who extract data, and methodologists who need dual review, adjudication, and inter-rater statistics for Cochrane-style protocols (Higgins et al., 2023). A secondary audience is NLP researchers, for whom the per-field decision chain (AI pre-fill, R1, R2, adjudicator) is a reusable data asset.

5 Related Work

Systems for evidence synthesis. AUTOFOREST (Pronesti et al., 2026) targets a different endpoint (Cochrane forest-plot generation from PDFs) with interactive correction of ICO elements, extracted values, and synthesis outputs. SCHEMATIQ (Levy et al., 2026) discovers a schema from a research question plus a document collection and supports expert revision of the schema and the extracted table; it is our closest general-purpose sibling.

EVISTREAMS instead assumes a protocol-defined extraction form and focuses on controlled execution, inserting review at *three* key stages (the extraction plan, approved before deterministic code generation; the field specification, via pilot calibration; and the predictions, via reviewer-blinded dual review with adjudication), with the structured field specification, not a free-text prompt, as the control surface. This three-level structure mirrors, rather than replaces, the dual-extraction-and-adjudication workflow review teams already run.

ROBOTO2 (Hevia et al., 2025) assesses risk of bias over a fixed schema; TRIALSTREAMER (Nye et al., 2020), CLINICALTRIALSHUB (Park et al., 2026), SCICLAIMS (Ortega and Gomez-Perez, 2025), and ASREVIEW (van de Schoot et al., 2021) automate adjacent slices. Established review tools (Marshall et al., 2016; Sun et al., 2025) and annotation platforms (Pei et al., 2022; Klie et al., 2018) place the human loop at the prediction level only, correcting outputs rather than the program that produces them.

Specification, decomposition, and data extraction. Our thesis turns a growing observation (that objective articulation can matter more than model choice (Hein et al., 2025; Li et al., 2025)) into a structured, editable control surface tested across models. Concurrent clinical-extraction work independently corroborates the robustness half of that finding (Aal Abdulsalam et al., 2026; Hart and Bergamaschi, 2026) (§3.1).

On the data side, VAXSCOPE (Ilgen et al., 2026) benchmarks review-level extraction with a fixed taxonomy, whereas EVISTREAMS exposes user-defined structured fields in a live loop. We adopt a variant of the error taxonomy of Simmons et al. (2025); the per-field extractor is a Chain-of-Thought module (Wei et al., 2022) compiled through DSPy (Khatab et al., 2024), and our anchor-column table extraction (§2.3) is a domain instance of decomposed prompting (Khot et al.,

2023; Liu et al., 2025).

We therefore do not claim novelty in pipeline decomposition or source-grounding in themselves (both appear in the systems above), but in unifying human review at three key stages over a structured, editable specification, and in showing empirically that, across our corpora, the field specification is the largest and most consistent control surface for quality.

6 Conclusion

EVISTREAMS is a human-in-the-loop platform for conducting systematic reviews via human–AI collaboration. It is made up of several parts: the extraction *program* (approved by users before it runs), the *specification* (iteratively refined by letting the user run test questions), and the *extractions* (validated through reviewer-blinded dual review). Our evaluation shows, across four clinical corpora and three frontier model families, that extraction quality is robust to model and pipeline decomposition, and that the structured field specification drives the biggest variance in quality. Expert effort is therefore best spent on the specification, with adjudication the second-most-useful place for human intervention to catch residual errors. Because a field’s specification can be edited and take effect immediately, without regenerating the pipeline, the lever our evaluation identifies as most valuable is also the cheapest one to pull. An extraction workflow that is expert-controlled and auditable is intended to reduce transcription work and supports systematic-review workflows.

Limitations

Ground truth and specification fit. When the ground truth was created, inter-annotator agreement was not measured, so we don’t know the human upper bound. Because field specifications were authored and calibrated against these same corpora, part of the specification’s measured advantage may reflect fit to the evaluation data rather than a fully generalizable gain.

LLM-judge overlap. The free-text judge shares a provider family with one extraction model (§3), a potential circularity we do not yet quantify with a human-agreement rate.

Evidence tracking. Currently, only AI-extracted values retain provenance; a human-edited value loses its source location in the PDF.

Ethics Statement

EVISTREAMS is a research tool for evidence synthesis, not a medical device, and its outputs are not clinical advice; this work adheres to the ACL Code of Ethics and the ACM Code of Ethics and Professional Conduct. Its central design property is that no AI-extracted value enters a final dataset without human review: the dual-review and adjudication workflow (§2) enforces verification structurally rather than relying on user diligence.

The evaluation corpora are published, peer-reviewed articles; no patient-level or identifiable data is processed. Users upload PDFs they are licensed to access; uploaded text is sent to commercial LLM APIs (Anthropic, OpenAI, Google) subject to those providers' data-handling terms. The hosted instance restricts project data to invited team members via role-based access control.

Automated extraction errors could propagate into clinical guidance if left unchecked; we report the error analysis openly (§3.3), distinguishing genuine model errors from ground-truth and annotation-convention discrepancies, and the per-field decision chain records every step from AI suggestion to human-approved value.

References

- Abdulrahman Aal Abdulsalam, Adhari Al Zaabi, Riham Jeeballah, and Habiba El Keraby. 2026. [A multi-agent open-source LLM for structured cancer registry information extraction from pathology and medical reports](#). In *Proceedings of the 25th Workshop on Biomedical Language Processing (BioNLP)*, pages 531–551. Association for Computational Linguistics.
- Lakshya A Agrawal, Shangyin Tan, Dilara Soyulu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2025. [GEPA: Reflexive prompt evolution can outperform reinforcement learning](#). *Preprint*, arXiv:2507.19457.
- Anthropic. 2026. [Introducing Claude Sonnet 4.6](#).
- Romina Brignardello-Petersen, Alonso Carrasco-Labra, Ignacio Araya, Nicolas Yanine, Luis Cordova Jara, and Julio Villanueva. 2015. [Antibiotic prophylaxis for preventing infectious complications in orthognathic surgery](#). *Cochrane Database of Systematic Reviews*, (1):CD010266.
- Alonso Carrasco-Labra, Olivia Urquhart, Ankita Shashikant Bhosale, Francisca Verdugo-Paiva, Carolina Castro Martins-Pfeifer, Krishna Aravamudhan, Marko Vujicic, Tim Wright, and Michael Glick. 2026. [American Dental Association living evidence-informed guidelines: shaping the future of clinical and public oral health decision making](#). *The Journal of the American Dental Association*, 157(3):213–217.
- Datalab. 2024. Datalab: Document intelligence API. <https://www.datalab.to>.
- Google DeepMind. 2026. [Gemini 3.1 Pro model card](#).
- Steven N. Hart and Teya S. Bergamaschi. 2026. [Agent-based large language model system for extracting structured data from breast cancer synoptic reports: a dual-validation study](#). *JAMIA Open*, 9(1):ooag016.
- David Hein, Alana Christie, Michael Holcomb, Bingqing Xie, AJ Jain, Joseph Vento, Neil Rakheja, Ameer Hamza Shakur, Scott Christley, Lindsay G Cowell, James Brugarolas, Andrew R Jamieson, and Payal Kapur. 2025. [Iterative refinement and goal articulation to optimize large language models for clinical information extraction](#). *npj Digital Medicine*, 8:301.
- Anthony Hevia, Sanjana Chintalapati, Veronica Ka Wai Lai, Nguyen Thanh Tam, Wai-Tat Wong, Terry P Klassen, and Lucy Lu Wang. 2025. [ROBOTO2: An interactive system and dataset for LLM-assisted clinical trial risk of bias assessment](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 12–25, Suzhou, China. Association for Computational Linguistics.
- Julian P. T. Higgins, James Thomas, Jacqueline Chandler, Miranda Cumpston, Tianjing Li, Matthew J. Page, and Vivian A. Welch, editors. 2023. [Cochrane Handbook for Systematic Reviews of Interventions](#), version 6.4 (updated August 2023) edition. Cochrane.
- Bahar İlgen, Ebenezer Awotoloro, and Georges Hattab. 2026. [VaxScope: Document-level structured evidence extraction from immunization systematic reviews](#). In *Proceedings of the 25th Workshop on Biomedical Language Processing (BioNLP)*, pages 853–863. Association for Computational Linguistics.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. 2024. [DSPy: Compiling declarative language model calls into self-improving pipelines](#). In *International Conference on Learning Representations (ICLR)*.
- Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish Sabharwal. 2023. [Decomposed prompting: A modular approach for solving complex tasks](#). In *International Conference on Learning Representations (ICLR)*.

- Jan-Christoph Klie, Michael Bugert, Beto Boullosa, Richard Eckart de Castilho, and Iryna Gurevych. 2018. The INCEpTION platform: Machine-assisted and knowledge-oriented interactive annotation. In *Proceedings of the 27th International Conference on Computational Linguistics: System Demonstrations*, pages 5–9. Association for Computational Linguistics.
- LangChain. 2024. LangGraph: A framework for building stateful, multi-actor applications with LLMs. <https://langchain-ai.github.io/langgraph/>.
- Shahar Levy, Eliya Habba, Reshef Mintz, Barak Raveh, Renana Keydar, and Gabriel Stanovsky. 2026. *ScheMatiQ: From research question to structured data through interactive schema discovery*. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 220–230. Association for Computational Linguistics.
- Lingbo Li, Anuradha Mathrani, and Teo Susnjak. 2025. What level of automation is “good enough”? A benchmark of large language models for meta-analysis data extraction. *Preprint*, arXiv:2507.15152.
- Shuodi Liu, Yingzhuo Liu, Zi Wang, Yusheng Wang, Huijia Wu, Liuyu Xiang, and Zhaofeng He. 2025. Select-then-decompose: From empirical analysis to adaptive selection strategy for task decomposition in large language models. *Preprint*, arXiv:2510.17922.
- Iain J Marshall, Joël Kuiper, and Byron C Wallace. 2016. RobotReviewer: Evaluation of a system for automatically assessing bias in clinical trials. *Journal of the American Medical Informatics Association*, 23(1):193–201.
- Benjamin Nye, Ani Nenkova, Iain Marshall, and Byron C. Wallace. 2020. Trialstreamer: Mapping and browsing medical evidence in real-time. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 63–69. Association for Computational Linguistics.
- OpenAI. 2026. *Introducing GPT-5.5*.
- Krista Opsahl-Ong, Arnav Singhvi, Dilara Soyly, Omar Khattab, Matei Zaharia, and Christopher Potts. 2024. *MIPROv2: Optimizing instructions and demonstrations for multi-stage language model programs*. *Preprint*, arXiv:2406.11695.
- Raúl Ortega and Jose Manuel Gomez-Perez. 2025. *Sci-Claims: An end-to-end generative system for biomedical claim analysis*. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 141–154, Suzhou, China. Association for Computational Linguistics.
- Jiwoo Park, Ruoqi Liu, Avani Jagdale, Andrew Srisuwananukorn, Jing Zhao, Lang Li, Ping Zhang, and Sachin Kumar. 2026. *ClinicalTrialsHub: Bridging registries and literature for comprehensive clinical trial access*. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Rabat, Morocco. Association for Computational Linguistics.
- Jiaxin Pei, Aparna Ananthasubramaniam, Xingyao Wang, Naitian Zhou, Apostolos Dedeloudis, Jackson Sargent, and David Jurgens. 2022. *POTATO: The portable text annotation tool*. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 327–337. Association for Computational Linguistics.
- Sara Pessano, Natasha R. Gloeck, Luca Tancredi, Martin Ringsten, Ameer Hohlfeld, Sumayyah Ebrahim, Martina Albertella, Tamara Kredo, and Matteo Bruschetti. 2024. *Ibuprofen for acute postoperative pain in children*. *Cochrane Database of Systematic Reviews*, (1):CD015432.
- Massimiliano Pronesti, Angelo Miculescu, Mohsin Kapdi, Paul Flanagan, Oisín Redmond, Joao Bettencourt-Silva, Gurdeep S. Mannu, Spiros Denaxas, Rui Bebiano Da Providencia E Costa, Anya Belz, and Yufang Hou. 2026. *AutoForest: Automatically generating forest plots from biomedical studies with end-to-end evidence extraction and synthesis*. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 128–137. Association for Computational Linguistics.
- Zalaya Simmons, Beti Evans, Tamsyn Harris, Harry Woolnough, Lauren Dunn, Jonathon Fuller, Kerry Cella, and Daphne Duval. 2025. *Assessing the feasibility and acceptability of a bespoke large language model pipeline to extract data from different study designs for public health evidence reviews*. *Cochrane Evidence Synthesis and Methods*, 3(6):e70061.
- Terry C. Simpson, Jan E. Clarkson, Helen V. Worthington, Laura MacDonald, Jo C. Weldon, Ian Needleman, Zipporah Iheozor-Ejiofor, Sarah H. Wild, Aisha Qureshi, Alice Walker, Veena A. Patel, Dwayne Boyers, and Joshua Twigg. 2022. *Treatment of periodontitis for glycaemic control in people with diabetes mellitus*. *Cochrane Database of Systematic Reviews*, (4):CD004714.
- Qiang Sun, Sirui Li, Tingting Bi, Du Q Huynh, Mark Reynolds, Yuanyi Luo, and Wei Liu. 2025. DocSpiral: A platform for integrated assistive document annotation through human-in-the-spiral. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*.
- Rens van de Schoot, Jonathan de Bruin, Raoul Schram, Parisa Zahedi, Jan de Boer, Felix Weijdemans, Bianca Kramer, Martijn Huijts, Maarten Hoogerwerf, Gerbrich Ferdinands, Albert Harkema, Joukje Willemsen,

Yongchao Ma, Qixiang Fang, Sybren Hindriks, Lars Tummers, and Daniel L Oberski. 2021. [An open source machine learning framework for efficient and transparent systematic reviews](#). *Nature Machine Intelligence*, 3:125–133.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

A Ablation and Model Coverage

All three model families (Claude, GPT, Gemini) were run through the production pipeline, the single-prompt arm (decomposition ablation), and the description-only arm (specification ablation) on all four corpora.

B Per-Corpus Production Results

Table 3 gives the per-form macro-F1 for all four corpora behind the main-text corpus-level micro-F1 (Table 1). Interventions and outcomes are record-level (one row per arm, or per arm $\times$ time point).

Corpus	Form	Gemini	GPT	Claude
Ibuprofen	study_char.	0.790	0.827	0.829
	patient_pop.	0.920	0.938	0.933
	interventions	0.892	0.900	0.915
	continuous_out.	0.795	0.772	0.765
	dichotomous_out.	0.774	0.754	0.792
	Overall	0.834	0.838	0.847
Oral cancer	study_char.	0.721	0.736	0.730
	patient_pop.	0.788	0.788	0.819
	reference_std.	0.682	0.687	0.695
	index_test	0.679	0.721	0.724
	Overall	0.718	0.733	0.742
Antibiotic	study_char.	0.921	0.907	0.919
	patient_pop.	0.862	0.880	0.880
	interventions	0.920	0.853	0.922
	dichotomous_out.	0.874	0.746	0.862
	Overall	0.894	0.847	0.896
Periodontitis	study_char.	0.878	0.802	0.859
	patient_pop.	0.777	0.787	0.815
	interventions	0.819	0.790	0.881
	continuous_out.	0.830	0.807	0.822
	Overall	0.826	0.797	0.844

Table 3: Per-form macro-F1, three model families through the same pipeline with identical form specifications, across all four corpora (best per row in bold; ties bolded on both). Corpus-level micro-F1 is in Table 1. Periodontitis is scored on the canonical 29-field set (OPTIONAL and structural row-matching fields excluded).

C Specification Ablation, Full Detail

Table 4 gives the per-form specification ablation ($\Delta F_1 = \text{full} - \text{description-only}$) for all three model families, the finding summarized in §3.2. Full-pipeline per-form macro-F1 is in Table 3 (Appendix B). The oral-cancer index_test recall collapse discussed in the main text reproduces across all three models (Claude +0.476, Gemini +0.489, GPT +0.433).

Corpus	Form	Cld.	Gem.	GPT
Antibiotic	study_char.	+0.012	+0.029	+0.014
	patient_pop.	0.000	−0.018	+0.031
	interventions	+0.029	+0.026	+0.005
	dichotomous_out.	+0.014	+0.037	+0.092
Periodont.	study_char.	−0.027	+0.013	+0.036
	patient_pop.	+0.014	+0.002	+0.008
	interventions	+0.065	+0.116	+0.307
	continuous_out.	+0.004	+0.022	+0.058
Ibuprofen	study_char.	−0.011	−0.030	+0.010
	patient_pop.	−0.001	+0.002	+0.010
	interventions	−0.014	+0.041	+0.018
	continuous_out.	−0.033	+0.001	+0.006
	dichotomous_out.	+0.044	0.000	−0.050
Oral canc.	study_char.	−0.010	+0.015	+0.018
	patient_pop.	+0.048	+0.029	+0.005
	reference_std.	+0.049	+0.015	+0.071
	index_test	+0.476	+0.489	+0.433

Table 4: Specification ablation, $\Delta F_1 = \text{full} - \text{description-only}$, macro-F1, per form across all three model families. Positive favors the full specification; bold marks the oral-cancer index_test collapse, reproduced by all three models.

D Decomposition Ablation

Table 5 gives the full corpus-by-modality breakdown (table vs. scalar forms, averaged over all three model families) behind the robustness finding in §3.1.

Corpus	Table ΔF_1	Scalar ΔF_1
Oral cancer	+0.027	+0.003
Antibiotic	−0.036	−0.012
Periodontitis	−0.002	−0.012
Ibuprofen	0.000	+0.003

Table 5: Decomposition ablation: overall ΔF_1 (full multi-signature pipeline minus single-prompt arm), split by table and scalar forms and averaged over Claude, GPT, and Gemini. Positive favors the full pipeline.

E Detailed Error Analysis

Table 6 expands the taxonomy in Table 2, giving for each discrepancy type what it is and why it occurs.

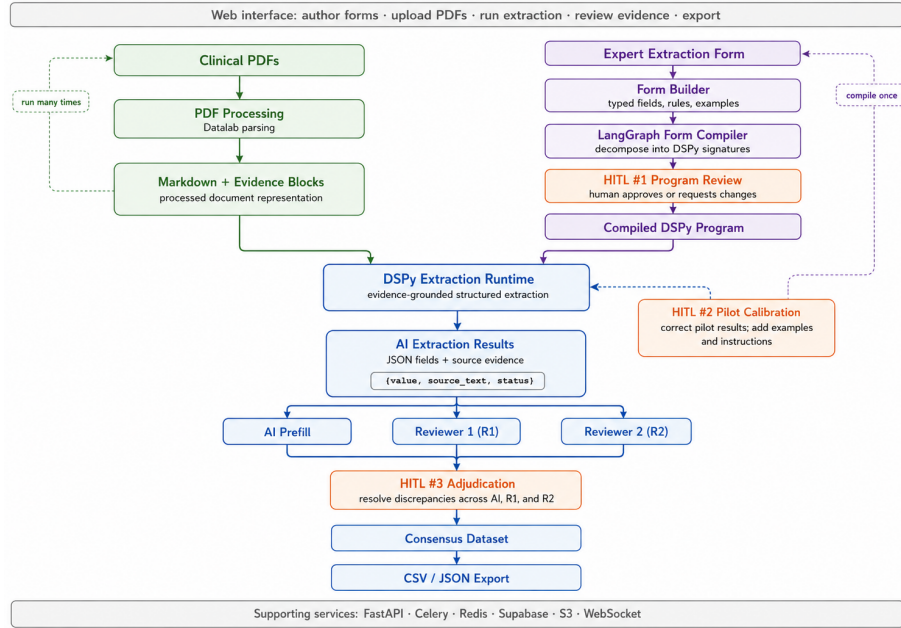


Figure 5: The EVISTREAMS pipeline. Human-in-the-loop (HITL) review enters at three key stages: the extraction *program* (HITL #1), its *specification* (HITL #2), and its *predictions* (HITL #3).

Discrepancy type	What it is	Why it occurs
Methodological evaluation-side 683/1003 (68%)	or Four patterns: the ground-truth cell is wrong, blank, or falsely coded NR (27%); a report-silent reviewer convention (back-calculated SD, converted dose, figure read off a graph) vs. the literal value (24%); a field with two defensible readings and no multi-arm tie-break (11%); or a surface-form mismatch, e.g. explained vs. bare NR (6%).	Ground truth carries transcription slips and undocumented conventions the extraction is deliberately not allowed to infer; specifications do not disambiguate every multi-arm edge case; and scoring does not normalize every equivalent string.
Genuine model errors 298/1003 (30%)		
Missing record or outcome row 75/1003 (7%)	A whole row that should appear in a multi-arm outcome table (a subgroup, a secondary time point, or a prose-only outcome) is absent, not merely mis-valued.	Dense multi-arm, multi-time-point tables (and prose-only outcomes) are harder for row enumeration than single-value fields.
Stated-value omission (false NR) 63/1003 (6%)	NR was coded for an event count or arm size stated plainly in the report text or a results-table cell, needing no calculation.	The value sits inside dense multi-arm results prose or a table cell the extraction did not check before defaulting to NR.
Wrong arm, table, or time point 60/1003 (6%)	A value was produced but taken from the wrong arm, wrong table, or wrong time-point bin (or a plain transcription slip).	Multiple similar-looking tables, subgroups, and time points sit close together in these results sections.
Cohort or adjacent-column confusion 49/1003 (5%)	A value taken from the whole-trial or a nested sub-cohort N instead of the arm's analysed N, or from an adjacent time-point column.	Multi-stage randomization and adjacent per-time-point columns in dense baseline tables invite cohort and column mix-ups.
Denominator or document-context error 37/1003 (4%)	or A stated value was dropped, the wrong patient/lesion denominator was assigned, or generic Discussion text was reported as the study's own Methods.	Diagnostic-accuracy papers report several similar counts (received vs. analysed, per-arm vs. per-technique); citation-heavy Discussion can read like procedure.
Arm or regimen conflation 14/1003 (1%)	The wrong arm's dose, regimen, or label was copied onto a cell (e.g. the placebo arm described with the active arm's schedule), or a stated fact was dropped.	Multi-arm drug-versus-placebo and short- versus long-course dosing descriptions sit close together in dense prose.
Uncertain/unadjudicable 22/1003 (2%)	A disagreement the reviewer could not confidently attribute to either a genuine model error or a methodological/ground-truth issue from the paper text alone.	The source paper does not contain enough information to adjudicate the disagreement either way, so it is left unclassified rather than forced into a category.

Table 6: Detailed error definitions and causes for the discrepancy types in Table 2 (counts and shares are given there).